Lecture (4):CHEMICAL BONDING

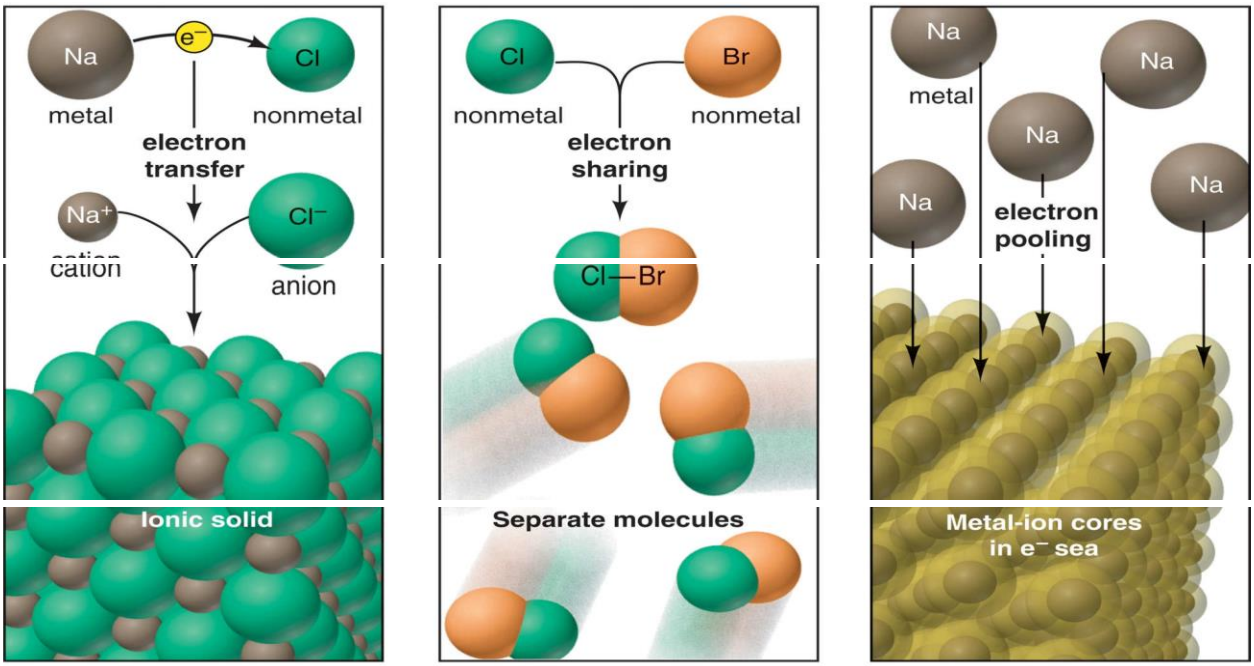
Chemical bonding is the creation of a chemical compound by forming a chemical link between two or more atoms, molecules, or ions. The atoms in the resulting molecule are held together by chemical bonds.

Types of chemical bonds include:

– **Covalent bonds** hold atoms together in molecules

– **Ionic bonds** hold ions together in ionic compounds

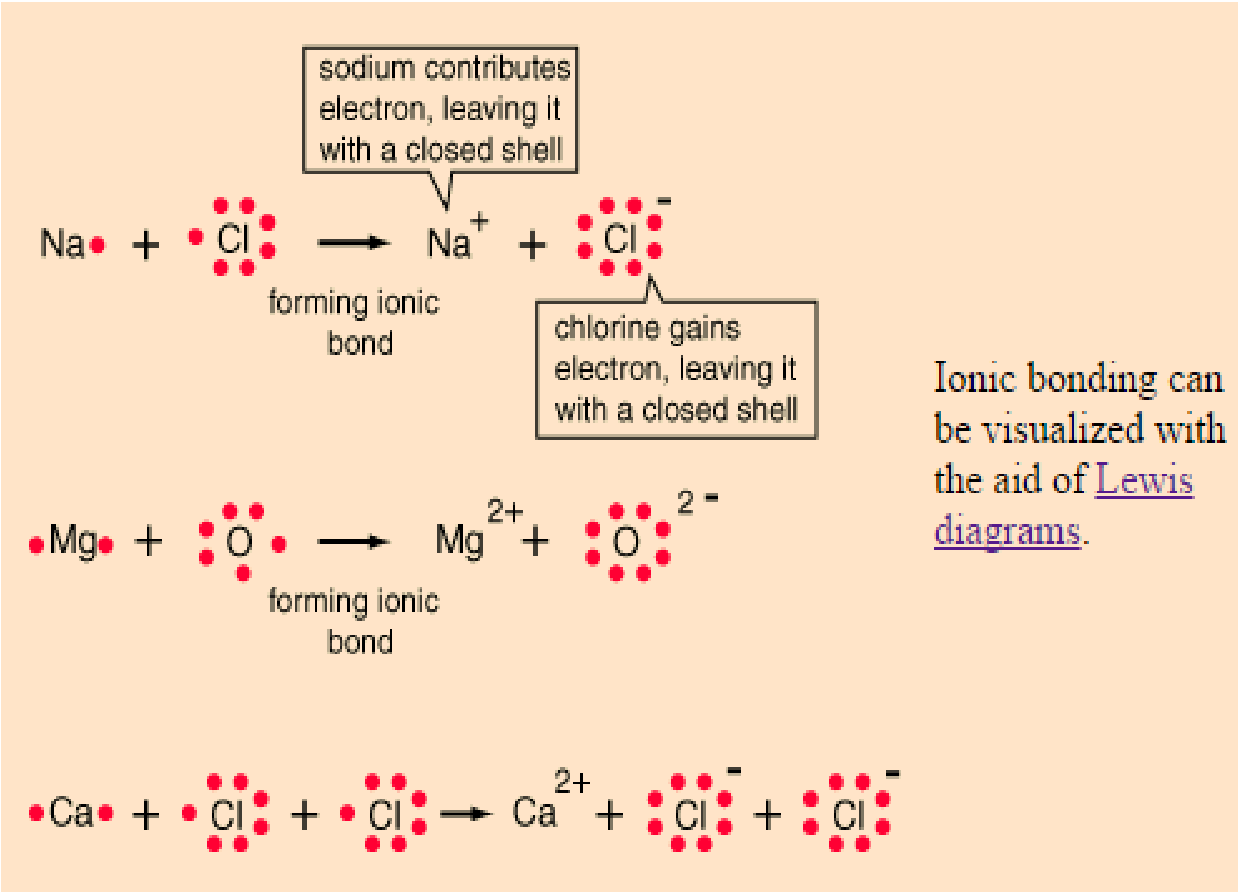
– **Metallic bonds** electron pooling and (delocalized).



A Ionic bonding B Covalent bonding C Metallic bonding

1- IONIC BOND:

An ionic bond is a chemical bond formed by the electrostatic attraction between positive and negative ions. The bond forms between two atoms when one or more electrons are transferred from the valence shell of one atom to the valence shell of the other. The atom that loses electrons becomes a cation (positive ion), and the atom that gains electrons becomes an anion (negative ion). As a result of the electron transfer, ions are formed, each of which has a noble-gas configuration. usually observed when a metal bonds to a nonmetal.

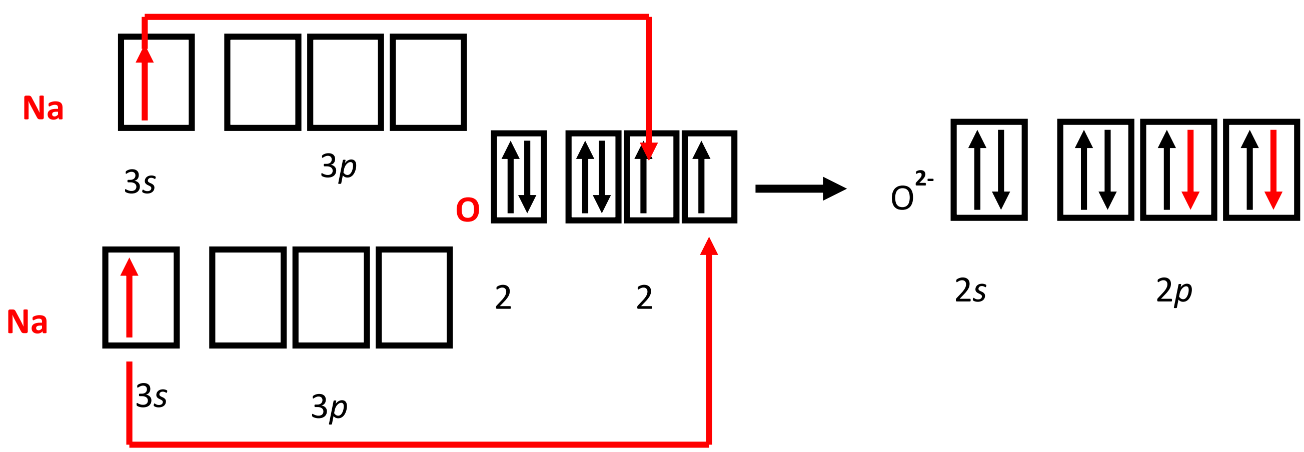


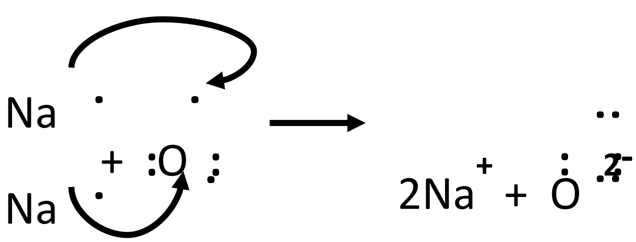
Example Depicting Ion Formation:

PROBLEM: Use partial orbital diagrams and Lewis symbols to depict the formation of Na^{+} and O^{2-} ions from the atoms, and determine the formula of the compound.

PLAN: Draw orbital diagrams for the atoms and then move electrons to make filled outer levels. It can be seen that two sodiums are needed for each oxygen.

SOLUTION:



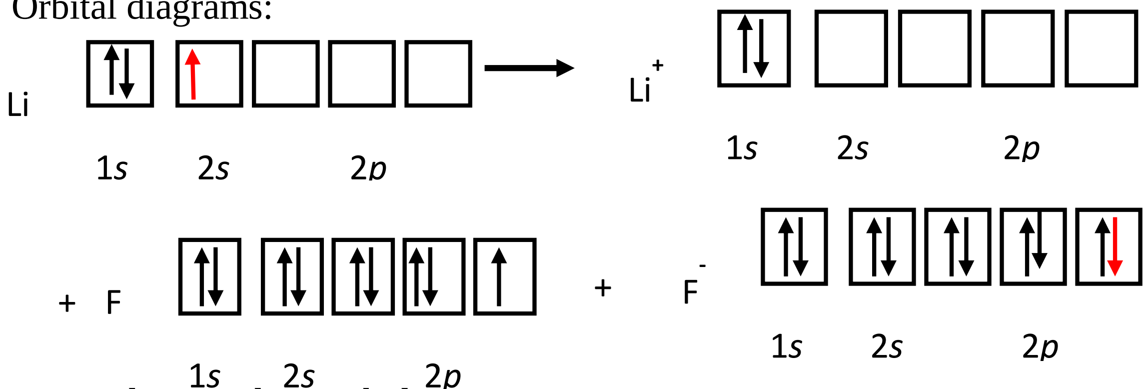


Three ways to represent the formation of Li^+ and F^- through electron transfer

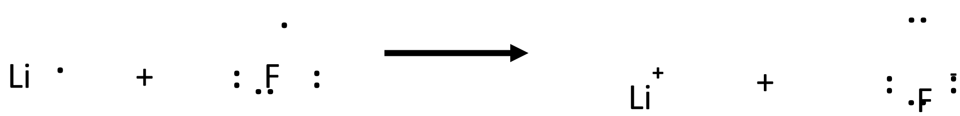
1. Electron configurations

$Li\ 1s^2 2s^1$ **+** $F\ 1s^2 2s^2 2p^5$ $\qquad Li^+\ 1s^2$ + $F^-\ 1s^2 2s^2 2p^6$

2. Orbital diagrams:



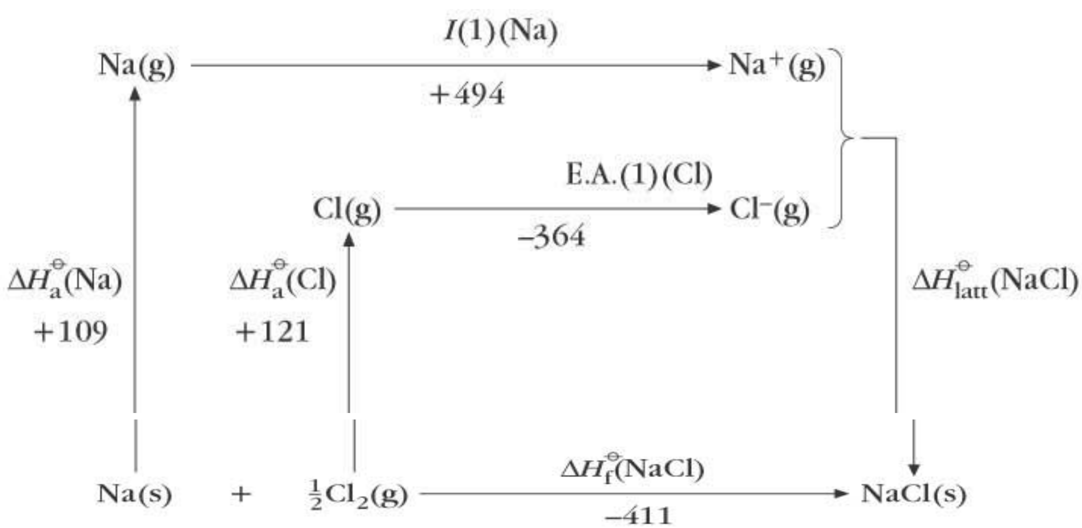
3. Lewis electron-dot symbols:



The Born-Haber cycle:

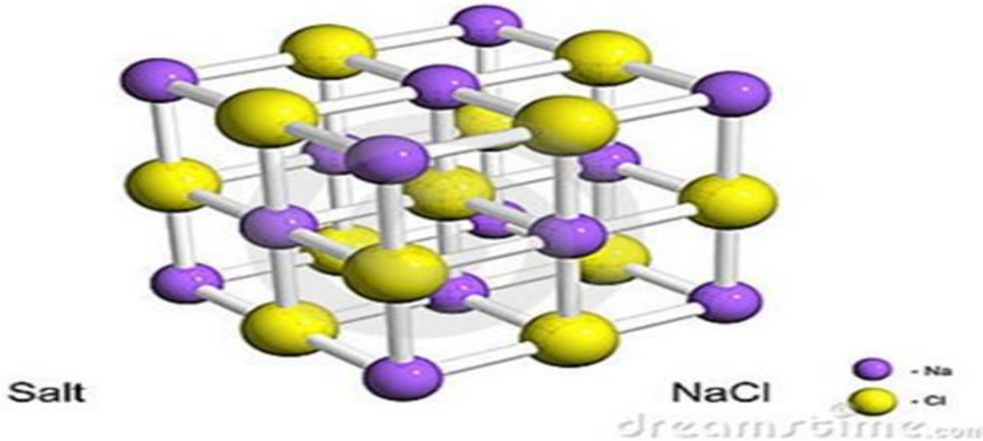
is a special case of Hess's law applied to ionic crystals only.

- Standard enthalpy of sublimation is the heat absorbed when one mole of atoms are vaporized.
- Standard enthalpy of bond dissociation is the energy absorbed when one mole of bonds is broken.
- The electron affinity is the energy absorbed when one mole of ions are formed from one mole of atoms.
- The standard lattice energy is the energy absorbed when one mole of ionic crystals is formed.



$-411 = 109 + 121 + 449 – 364 + \Delta Hl^{\ominus}$

Where $\Delta Hl^{\ominus}$ = standard lattice energy of NaCl = -726 KJ/mol



Factors governing the formation of ionic bonds:

(i) Ionisation Enthalpy (Ionization Energy)

(ii) Electron Affinity.

(iii) (iii) Lattice Energy

Ionic Bonding and Lattice Energy

The electron transfer process is an <u>endothermic</u> process, but ionic compound formation is an <u>exothermic</u> process.

- ➢ Lattice energy defined as the energy required to completely separate one mole of an ionic solid in an ionic compound into gaseous ions.

  Examples: NaCl $\longrightarrow$ Na+(g) + Cl-(g)

- ➢ Lattice energy cannot be measured directly but Lattice Energy can be measured indirectly by either.
- ➢ coloumb'slaw (using charges and radius)

  E $\propto q1q2r$ E = $14\pi\varepsilon 0q1q2r$

$$E = k \frac{q_1 \, q_2}{r}$$

q_1 is the charge on the cation, q_2 is the charge on the anion , r is the distance between the ions

	compound	Lattice energy	q_1,q_2	differentiate
Lattice energy (E) increases as q increases	MgF_2	2957	+2,-1	q of F<O
	MgO	3938	+2,-2	
Lattice energy (E) increases as as r decreases	LiF	1036	+1,-1	r F < r Cl
	LiCl	853	+1,-1	

or the Born-Haber cycle which depend on Hess's Law

- ➢ ($\Delta H^o_f = \sum \Delta H^o_1 + \Delta H^o_2 + \Delta H^o_3 +$ …………etc)

- (ΔHof= ∑ atomizatinn energy + ionization energy + electron affinity + bond energy + lattice energy………..etc).
- Therefore, we calculate the lattice energy from Hess's Law :

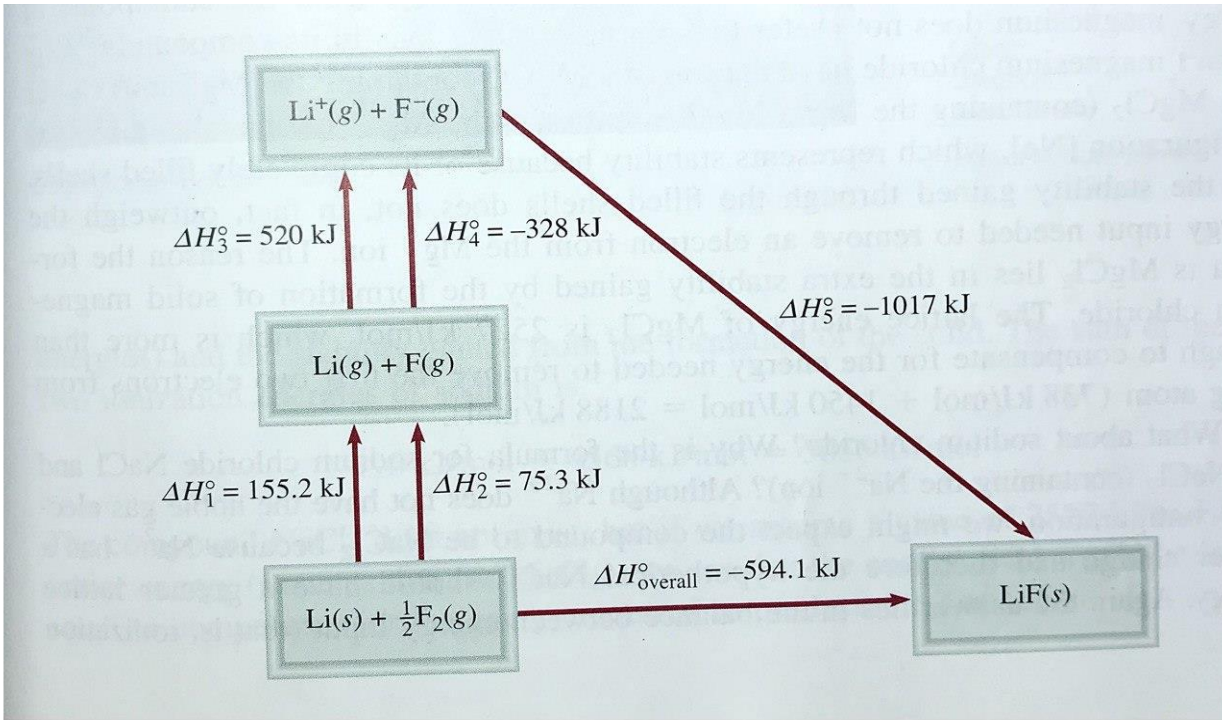
ΔH°f= -594.1 KJ = ΔH°step1 +ΔH°step2 + ΔH°step3 + ΔH°step4 + ΔH°LiF

Solving for ΔH°LiFgive:

= ΔH°f-(ΔH°step1 +ΔH°step2 + ΔH°step3 + ΔH°step4)

= -594.1 -[161 + 79.5 + 520 + (-328)]

= -1017 KJ



1.	$Li(s) \longrightarrow Li(g)$	$\Delta H_1^\circ = 155.2$ kJ/mol
2.	$\frac{1}{2}F_2(g) \longrightarrow F(g)$	$\Delta H_2^\circ = 75.3$ kJ/mol
3.	$Li(g) \longrightarrow Li^+(g) + e^-$	$\Delta H_3^\circ = 520$ kJ/mol
4.	$F(g) + e^- \longrightarrow F^-(g)$	$\Delta H_4^\circ = -328$ kJ/mol
5.	$Li^+(g) + F^-(g) \longrightarrow LiF(s)$	$\Delta H_5^\circ = ?$
	$Li(s) + \frac{1}{2}F_2(g) \longrightarrow LiF(s)$	$\Delta H^\circ_{overall} = -594.1$ kJ/mol

(an exothermic process due to the attraction of oppositely charged ions).

Compound	Lattice Energy (kJ/mol)	Melting Point (°C)
LiF	1017	845
LiCl	828	610
LiBr	787	550
LiI	732	450
NaCl	788	801
NaBr	736	750
NaI	686	662
KCl	699	772
KBr	689	735
KI	632	680
$MgCl_2$	2527	714
Na_2O	2570	Sub*
MgO	3890	2800

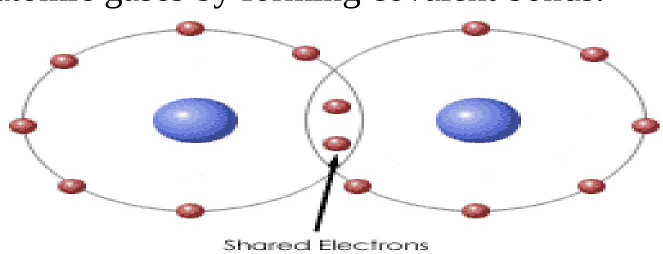
*Na_2O sublimes at 1275°C.

Characteristics of Ionic Compounds: They are solids with high melting points (typically >400°C).

1. Many are soluble in polar solvents such as water.
2. Most are insoluble in nonpolar solvents, such as hexane, C_6H1_4, and carbon tetrachloride, CCl_4.
3. Molten compounds conduct electricity well because they contain mobile charged particles (ions).
4. Aqueous solutions conduct electricity well because they contain mobile charged particles (ions).
5. They are often formed between two elements with quite different electronegativities, usually a metal and a nonmetal.

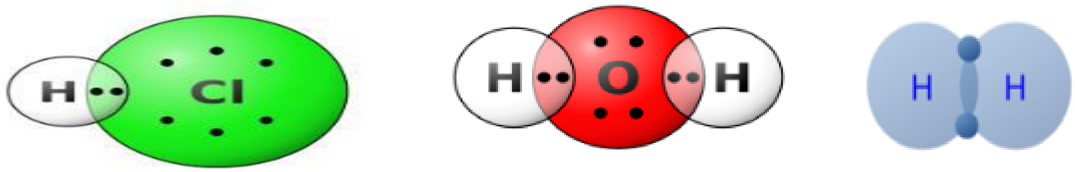
2- **Covalent bond:** a chemical bond formed by the sharing of a pair of electrons between atoms.is usually observed when a nonmetal bonds to a nonmetal.

Covalent bonding can be visualized with the aid of Lewis diagrams. The halogens such as chlorine also exist as diatomic gases by forming covalent bonds.



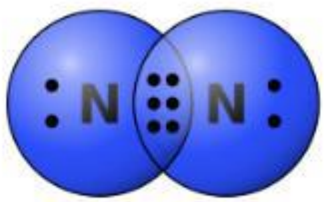
Types of covalent bond:

a **single bond***:* That is*, a covalent bond in which a single pair of electrons is shared by two atom such as* H_2 *,*Cl_2 *,*



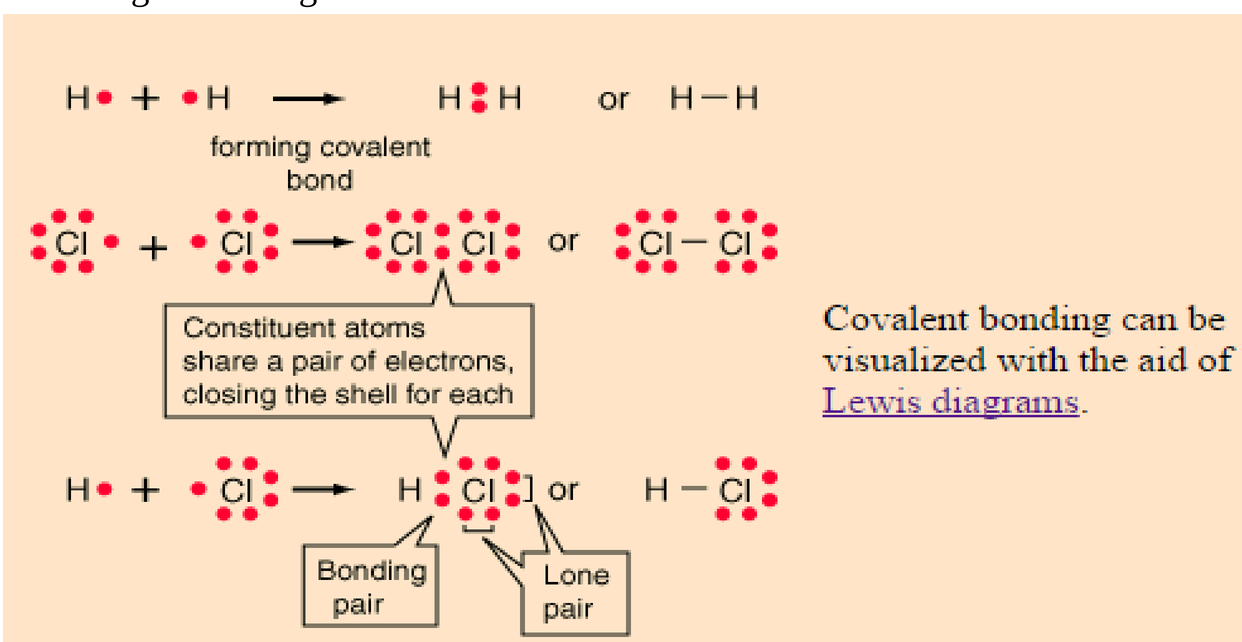
A **double bond** is *a covalent bond in which two pairs of electrons are shared by two atoms* consider ethylene, C_2H_4, O_2.

A **triple bond** is *a covalent bond in which three pairs of electrons are shared by two atoms.* As examples,Nitrogen and acetylene, C_2H_2.

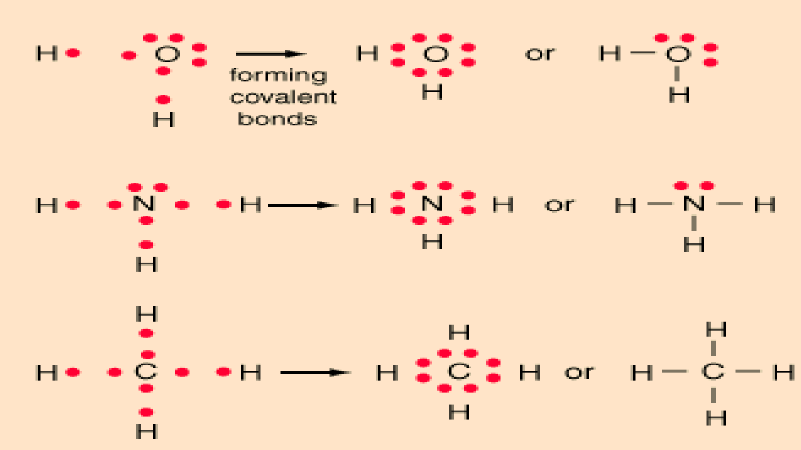


Hydrogen gas forms the simplest covalent bond in the diatomic hydrogen molecule.

. The nitrogen and oxygen which makes up the bulk of the atmosphere also exhibits covalent bonding in forming diatomic molecules.



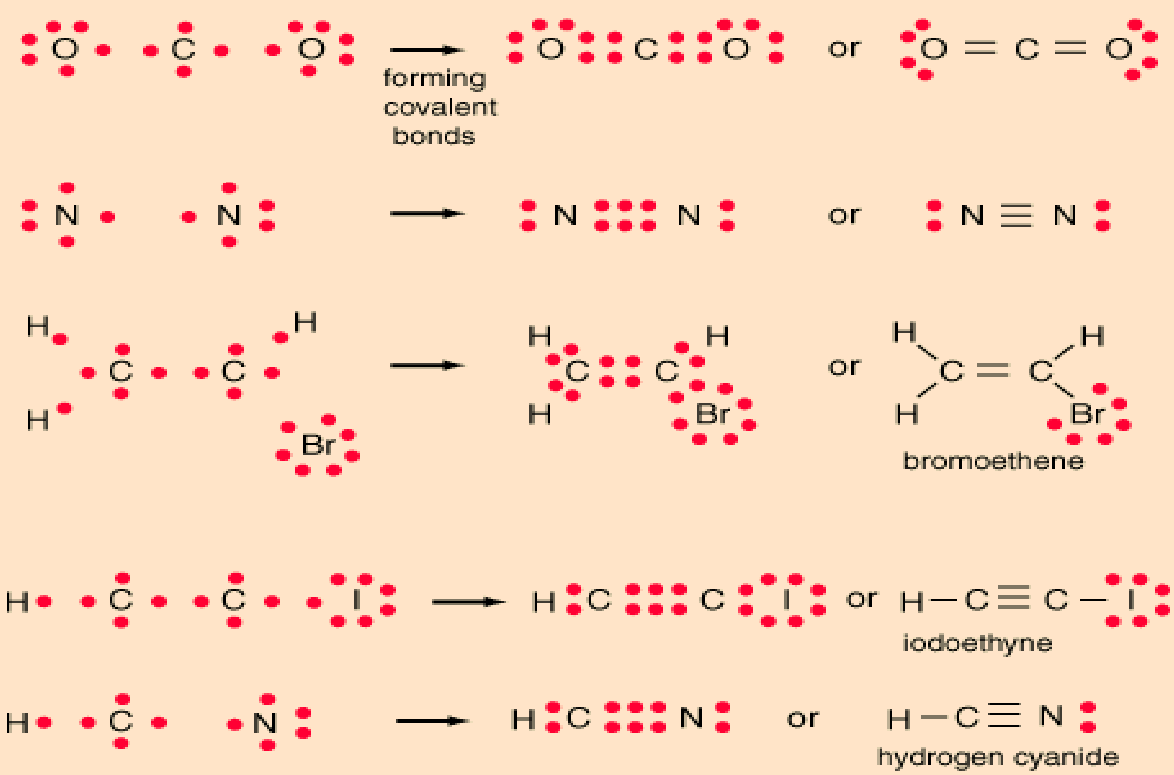
Lewis symbols and Lewis diagrams can be used to describe multiple bonds, but further information must be supplied to account for the three dimensional geometry of the resulting molecules. For multiple single bonds, the procedure is similar that for a single bond.



A single bond can be represented by the two dots of the bonding pair, or by a single line which represents that pair. The single line representation for a bond is commonly used in drawing Lewis structures for molecules.

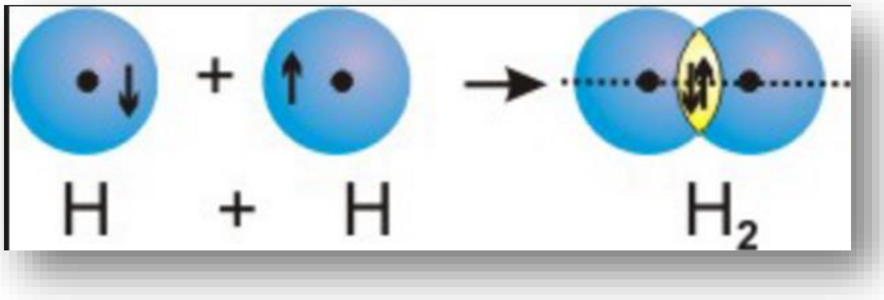
The Lewis structures are useful for visualization, but do not reveal the bent structure for water (105°), the pyramidal shape for ammonia, or the tetrahedral geometry of the methane molecule.

The Lewis diagrams can also help visualize double and triple bonds



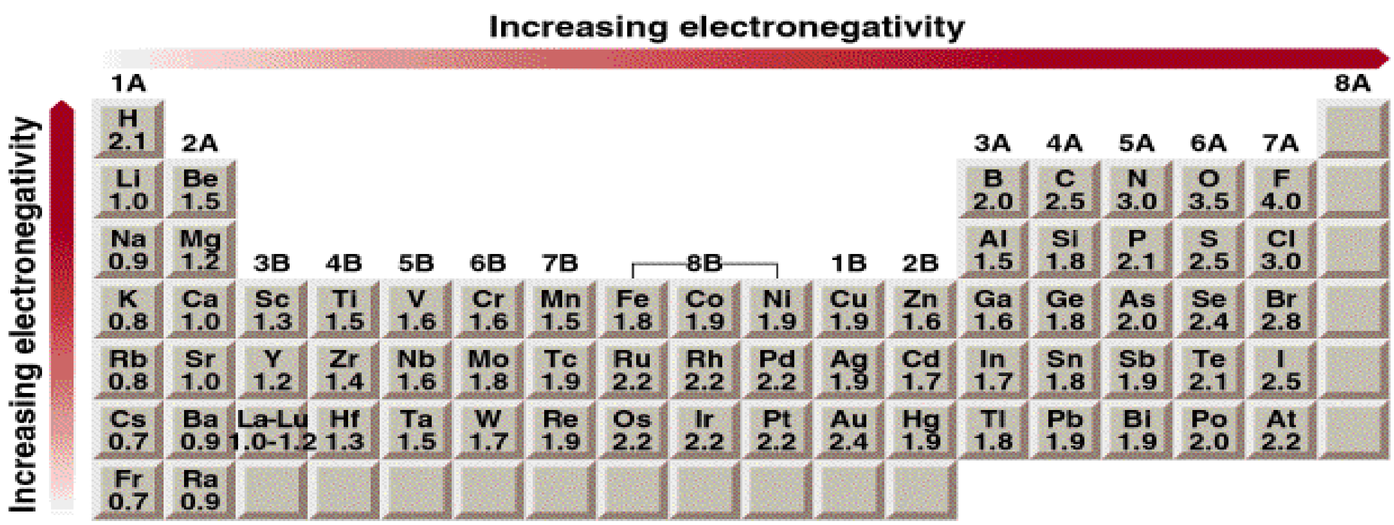
Nonpolar Covalent Bond

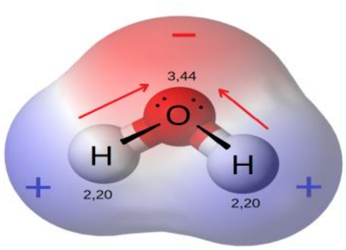
- In some covalent bonds, both atoms have equal electronegativity values or different atoms which have identical electronegativity values.
- The two atoms share the bonding electrons equally.
- Most common example is between two identical atoms: H_2, O_2, N_2, Cl_2, F_2, I_2, Br_2

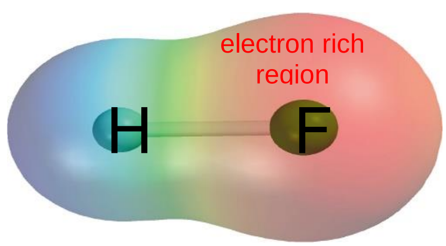


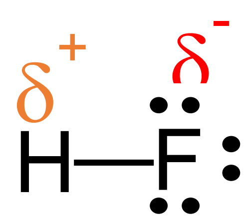
Polar Covalent Bonds

- In such a bond there is a charge separation with one atom being slightly more positive and the other more negative, i.e., the bond will produce a dipole moment.
- The ability of an atom to attract electrons in the presence of another atom is a measurable property called electronegativity.
- **Electronegativity** is *a measure of the ability of an atom in a molecule to draw bonding electrons to itself.*









Bond Parameters:

Bond length (or **bond distance**) is *the distance between the nuclei in a bond.* Bond lengths are determined experimentally using x-ray diffraction or the analysis of molecular spectra.

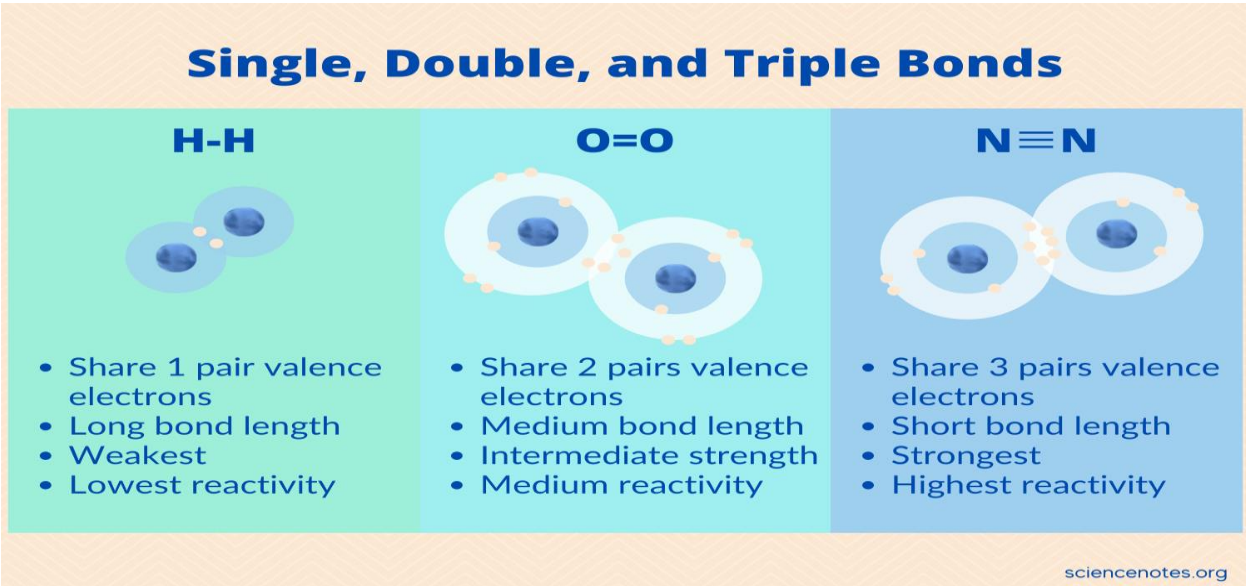
In many cases, bond lengths for covalent single bonds in compounds can be predicted from covalent radii.

Covalent radii are *values assigned to atoms in such a way that the sum* *of covalent radii of atoms A and B predicts an approximate A-B bond length*

Bond order :in terms of the Lewis formula, is *the number of pairs of electrons in a bond*. For example, in (C- C) the bond order is 1 (single bond); in(C = C) the bond order is 2 (double bond).

Bond energy: The amount of energy required to break one mole of bonds of a particular type so as to separate them into gaseous atoms is called bond dissociation enthalpy or simply bond energy.

Bond angle: The angle between the lines representing the directions of the bonds, i.e., the orbitals containing the bonding electrons is called the bond angle.



.

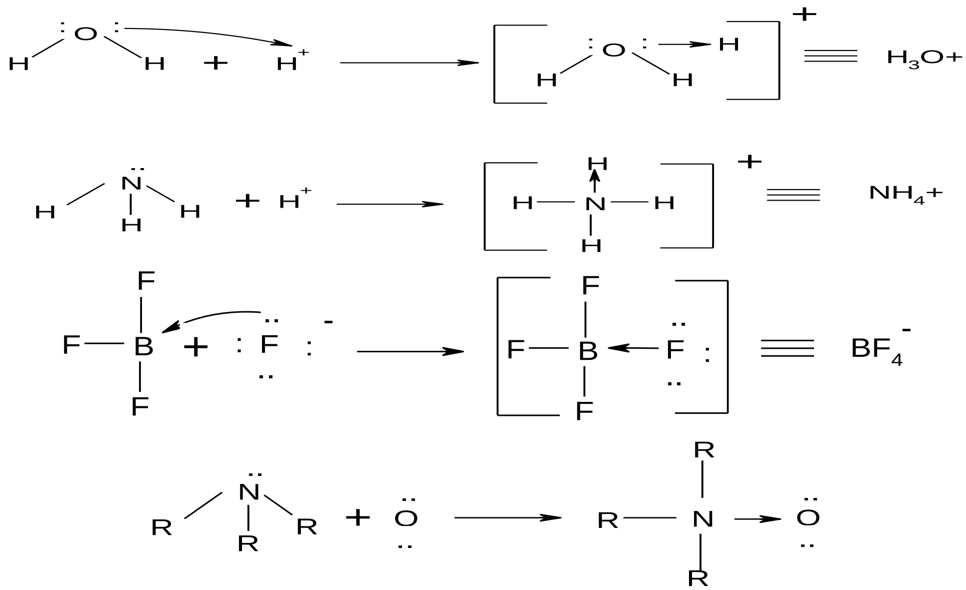
The properties of covalent compounds:

1. They can be found in three states.:solids such as I_2, liquids such as $Br_2$and gases such as F_2, Cl_2, H_2, N_2, O_2, ……etc. some solids are brittle and hard, but many are softs.
2. with low melting points (typically <300°C). and low boiling points.
3. Many are insoluble in polar solvents such as water utMost are soluble in nonpolar solvents.
4. Liquid and molten compounds do not conduct electricity.
5. Aqueous solutions are *usually* poor conductors of electricity because most do not contain charged particles
6. They are often formed between two elements with similar electronega tivities, usually nonmetals.

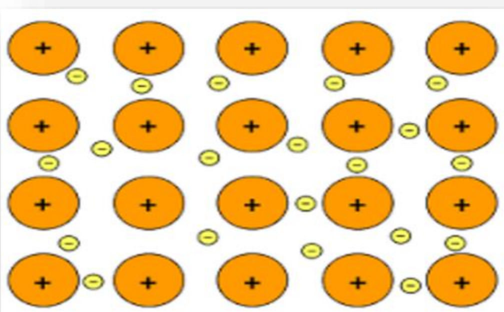
Coordinate bond(dative bond):

- It is a covalent bond in which the shared electron pair come from one atom is called coordinate.
- Necessary conditions for the fomation of co-ordinate bond are:
  (a) Octet of donor atom should be complete and should have atleast one lone pair of clectron.
  (b) Acceptor atom should have a deficiency of at least one pair of electron.
- Atom which provide clectron pair for sharing is callcd donor.

- Other atom which accepts electron pair is called acceptor. That is why it is called donor-acceptor or dative bond



3- **Metallic Bonds** The properties of metals suggest that their atoms possess strong bonds, yet the ease of conduction of heat and electricity suggest that electrons can move freely in all directions in a metal.



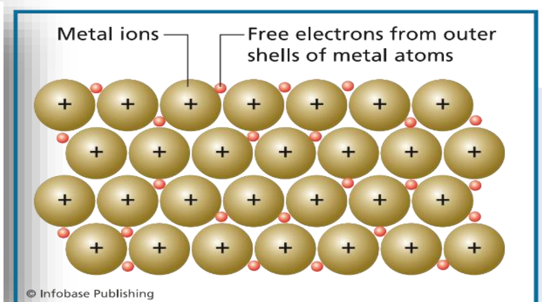
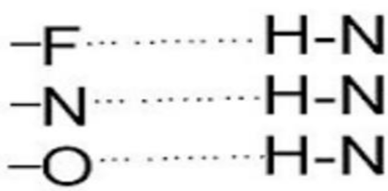


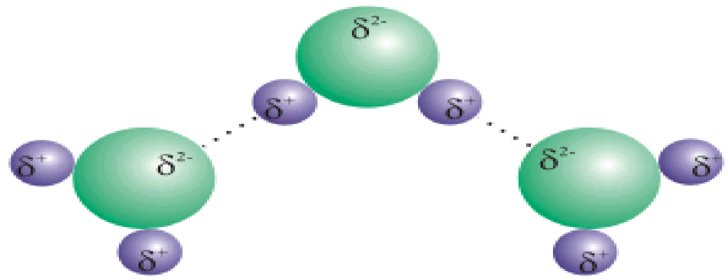
Figure 7.12
The outer electrons of metals are not bound to any one atom and easily move around in a sea of freely moving electrons.

element	mp (°C)	bp (°C)
lithium (Li)	180	1347
tin (Sn)	232	2623
aluminum (Al)	660	2467
barium (Ba)	727	1850
silver (Ag)	961	2155
copper (Cu)	1083	2570
uranium (U)	1130	3930

INTERMOLECULAR ATTRACTION : These are weak attractive forces - much weaker than the strong electrostatic attraction is found in ionic and covalent bonds.

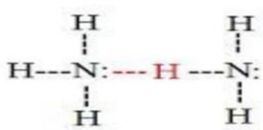
1- Hydrogen bonding: is the strongest of the intermolecular forces and seen as the particularly strong electrostatic interactions occurring between molecules of the type H—X. X is an electronegative atom such as F, O or N. O and N have also the advantage of possessing lone-pair orbitals. Hydrogen bonding can be seen as a strong and specialised form of dipole–dipole interaction and is the reason for the high boiling point of water.

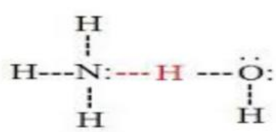


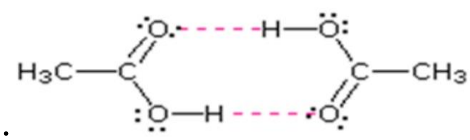


There is to type of hydrogen bond:

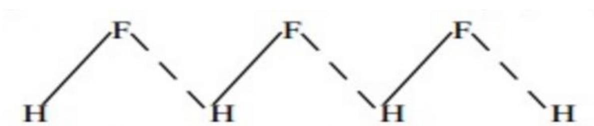
i. intermolecular (outer) hydrogen bond: confirmed between to molecules such as Hydrogen bond between water and ammonia And between ammonia and ammonia and two molecule of methanoic acid.



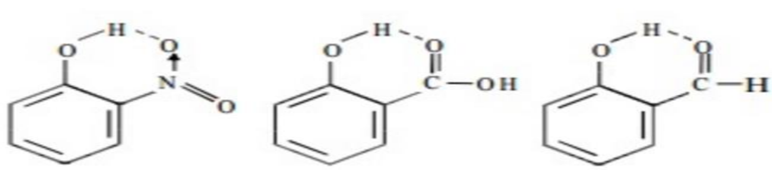




In the solid state, hydrogen fluoride consists of long zig-zag chains of molecules associated by hydrogen bonds



ii. Intramolecular hydrogen bonding: This type of bond is formed between hydrogen atom and N, O or F atom of the same molecule. Intramolecular hydrogen bonding is possible when a six or five membered rings can be formed.



O-Nitrophenol Salicylic Acid Salicylaldehyde

Strength of hydrogen bonds :

- Its strength depends upon the electronegativity of atom to which H atom is covalently bonded.
- Electronegativity of F > O > N . The strength of hydrogen bond is in the order of H-F - - - H > H-O --- H > H-N --- H .

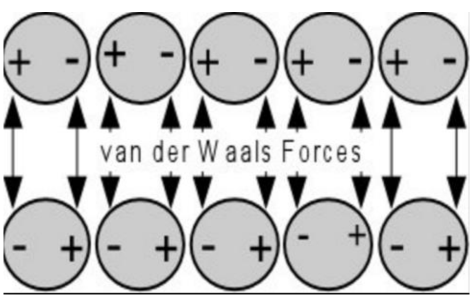
- Hydrogen bonds are much weaker than covalent bonds.

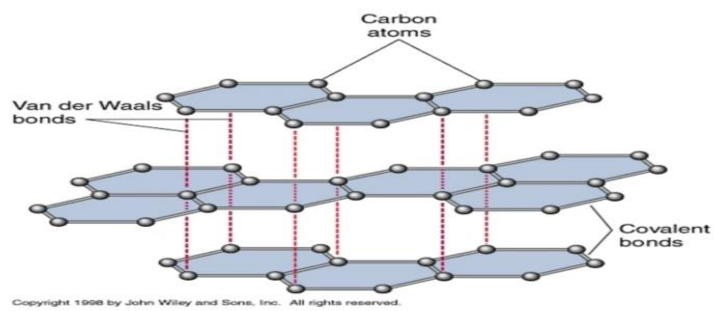
The bond strength of different bonds is in the order:

Ionic bond > Covalent bond > Hydrogen bond > dipole-dipole interaction, vander wall's (London forces).

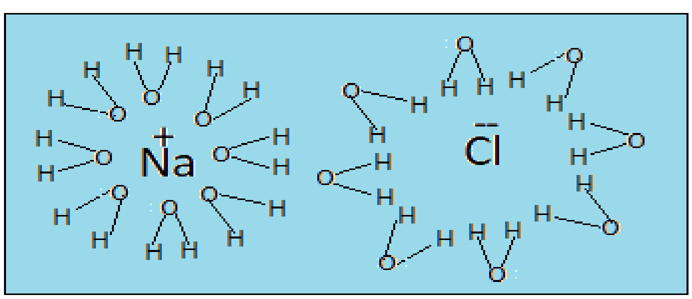
2- Vander Waals Bonding

- It is a weak bond, with a typical strength of 0.2 eV/atom.
- It occurs between neutral atoms and molecules.
- The explanation of these weak forces of attraction is that there are natural fluctuation in the electron density of all molecules and these cause small temporary dipoles within the molecules. It is these temporary dipoles that attract one molecule to another. They are called van der Waals' forces.
- The bigger a molecule is, the easier it is to polarise (to form a dipole), and so the van der Waal's forces get stronger, so bigger molecules exist as liquids or solids rather than gases.
- The **shape** of a molecule influences its ability to form temporary dipoles. Long thin molecules can pack closer to each other than molecules that are more spherical. The bigger the 'surface area' of a molecule, the greater the van der Waal's forces will be and the higher the melting and boiling points of the compound will be.
- Van der Waal's forces are of the order of 1% of the strength of a covalent bond.





3- Ion- dipole attraction - This force is between an ion such as and a polar molecule such as H_2O and an ionized salt NaCl as is shown below :-



Dipole -dipole interaction - It is again in between two polar molecules such as HF or HCl.